Hidden Markov Model: Statistical Analysis Steps

27/07/2026

Study Design

An HMM based on a retrospective longitudinal cohort of children born 2012-2019 with continuous residence in Lothian since birth.

Aim and objectives

To quantify the contribution of underheated homes to preschool ARI's by:

1. Developing the infrastructure for multilevel count models and latent state models.
2. Examining associations between home energy efficiency and ARI burden in 0 to 5-year-olds.

Exposure

Approach 1: At risk of living in a cold house

Use an initial definition of: EPC rating <= D and postcode level (median household) income is <60% of the Scottish median household income after housing costs (£34,258 based on 2022-2025 figures)[1]. This equates to: (EPC<=D) + (PC income <=£20,555), adjusted annually for inflation[2] - *Appendix 1.* It is the Low-Income, Low Energy Efficiency (LILEE) definition used to measure fuel poverty in England. Scotland and Wales use a different energy expenditure as a proportion of income definition (2M) - *Appendix 2*.

The advantage of this measure is that it acknowledges structural deficiencies in a house that may lead to it being cold, rather than just being based on affordability. However, household income (derived from the median for the data-zone) in Edinburgh is much higher than for the rest of Scotland, with only the lowest income decile in our cohort having an annual household income of approximately £35K and combining this with an EPC<=D, provides an exposed population of <2%.

Approach 2: Fuel Poverty measure:

2022-2024 Fuel Poverty Status, Scottish Housing Condition Survey						
% by area	East Lothian	Edinburgh	Midlothian	West Lothian	All Lothians	Scotland
Fuel poor	24.74	23.86	30.71	30.79	26.01	30.57
Not fuel poor	74.30	74.94	67.58	67.83	72.73	68.17
(blank)	0.96	1.20	1.71	1.38	1.26	1.26

Use the uZero derived *at risk of fuel poverty* score – with the highest quartile representing the exposure group (equal to 25% of records) - approximately equating to fuel poverty in The Lothians *(see above table)*. Risk of fuel poverty does not necessarily equate to *risk of living in a cold house,* although EPC rating is considered, alongside other variables – *Appendix 3*. A small number of records (1% of records, 1.1% of individuals) do not have a uZero Score (-99).

Approach 3: Relative Risk of Living in a Cold House

To acknowledge the influence of structural factors as well as income, a compromise may be to look at *relative risk of living in a cold house* defined as: (EPC<=D) + (PC income <=lowest tertile (approx. £49,000)). This provides an exposed population of approximately 10% in the Lothians, which is lower than and not

[1] https://ifs.org.uk/publications/labour-market-living-standards-and-poverty-trends-scotland
[2] https://www.bankofengland.co.uk/monetary-policy/inflation/inflation-calculator

generalisable to either the rest of Scotland or the UK. It also allows a crude four-way comparison to be made *(see table below).*

PC_income *EPC_rank	Tertile 1 (lowest)	Tertile 2-3 (med/high)	Total
ABC	Low Income + High Energy Efficiency	Med-high income + High Energy Efficiency	EPC=ABC (65%)
DEFG	Low Income + Low Energy Efficiency	Med-high income + Low Energy Efficiency	EPC=DEFG (35%)
Total	Low Income 30%	Med-high income 70%	Total child years (100%)

Exposure variable: Relative Risk of Living in a Cold House. Given the constraints of the data and focus on structural condition of the home in combination with energy affordability, the Lothian specific analysis will use approach 3 to define the exposure variable.

Outcome
Respiratory health state.

Covariates
Individual level: Sex, Ethnicity, Maternal smoking status, Comorbidities
Household level: Tenure, Floor area,
Small area level: Climate & Pollution (time-dependent), SIMD

Components

States (S)
1. Start with 3 states: Good Respiratory Health, Poor Respiratory Health, Very Poor Respiratory Health.
2. An individual's State (s_i) will be inferred from the observed variable/s (z_i) for each time point (t_j).

Observations (Z)
1. Prescriptions dispensed for respiratory illness (>= 1 per time point).
2. Hospital admissions for respiratory illness (>= 1 per time point) .
3. Mortality resulting from respiratory illness.

Time points (t)
1. Start point (t_0) aligns with birth.
2. Time points accrue (t_0 -> t_1 -> ….t_n) at one-month intervals.
3. End point (t_n) aligns with age five-years, end of the study period or death
4. Censoring time (t_c) aligns with the age at which an external event terminates the observation sequence (migration from Lothian or mortality). At t_c the individual is excluded from the dataset, based on the assumption that the censoring mechanism is independent of future states (*uninformed censoring*).

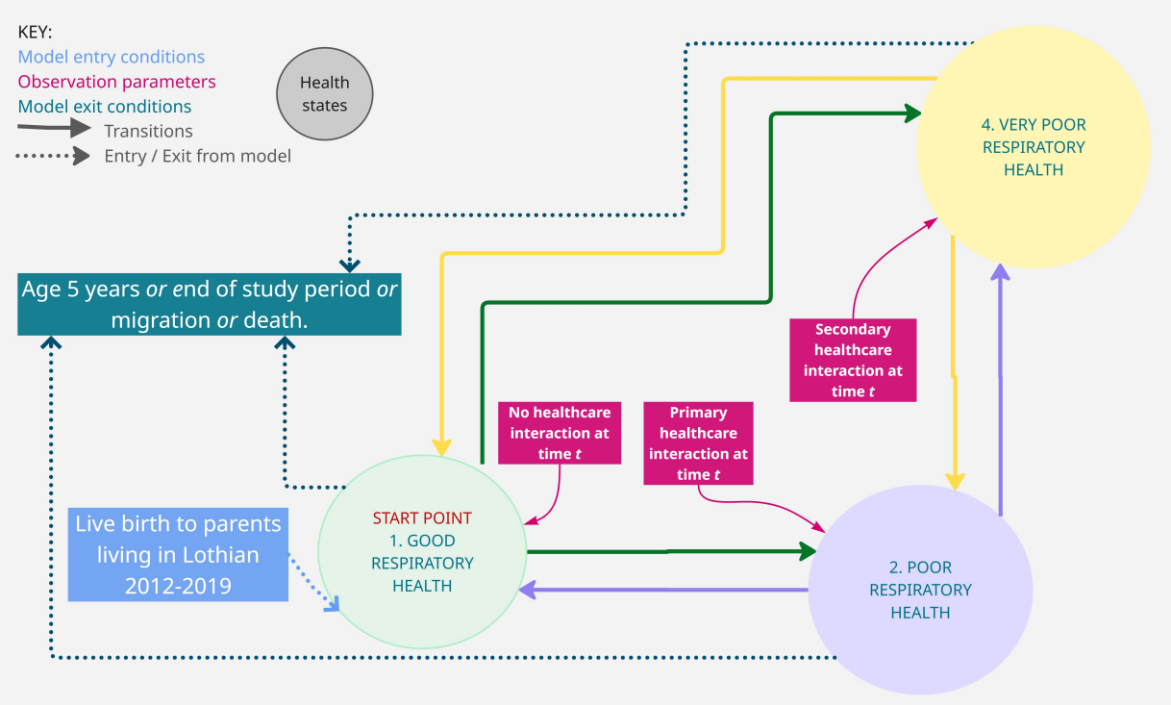
Transition probability Matrix (Q)
1. Transition probabilities will be based on the proportion of individuals in state (a) at t_j that transition to state (b) at t_{j+1}: i.e.: $P_{ab}=N_{ab}/N_a$.
2. Transitions probabilities will be derived from the emission distribution of (Z) at $t_{_j}$, assuming that these emissions are independent from observations during $t_{_\{j-1\}}$.

3. All possible transition probabilities, including remaining in the same state will be included in matrix Q (3x3 for a 3-state model).
4. The sum of all possible transitions from a state, a row in Q, i.e. (a->a, a->b, a->c), =1.

Random effects (R)

Random effects will be derived from covariates (listed above). Final selection will be based on a DAG to determine their relationship with the exposure and outcome, having taken collinearity into consideration. Priors will assume a zero mean normal distribution R_j~ Normal (0, σ_r^2), but with a wider dispersion.

Summary of components and their relationships



Data requirements

Repeat measures are required with one row for each time point. Individuals are differentiated using a UID. Likelihood of the full dataset is computed as the product of the likelihoods of all individuals (assuming independence).

Random effects should be formatted as factors – not character strings or integers

Missing values in response variables should be noted as NA.

Missing values in covariates need to be interpolated. The default, for NA values is to assign: (1) the last non-missing value, (2) the next non-missing value- where there are no previous values provided.

Analysis

Analysis will be conducted in R using the Bayesian inference package tmb_stan. Markov Chain Monte Carlo (MCMC) will be used to sample over random effects with (HMM$fit_stan()), to estimate their *conditional posterior distribution*, fitted using Hamiltonian Monte Carlo.

The model will be constructed from three classes:

1. **Markov Chain class** (MarkovChain$new()): consisting of the dataset (data), initial state, assuming healthy at birth (initial.state), the number of states - assume 3 (n.states), transition probability matrices

(tpm) and associated formulae (formulas) tpm formulae will be informed by a combination of raw Dataloch and PICNIC values.

2. **Observation class** (Observation$new()): consisting of observed variables (data), their distribution (dists)- for example normal, poisson etc. depending on how the observation has been classified, the number of states - assume 3 (n.states), formulae (formulas) where a variable has dependence on other covariates and initial parameter values (par) for variables.
3. **HMM class** (HMM$new()) combining the Markov Chain and Observational classes to perform statistical analysis.

Priors will be specified for fixed effects (α) and smoothness parameters (λ) (for splines) using (HMM$set.priors). The model will then be fitted (HMM$fit_stan()) – stating the number of chains and the number of iterations within each chain stated. These will be determined by numbers required to achieve convergence. This will be tested using a subset of the data, with a minimum starting point of 3 chains and 1,000 iterations.

Models will be constructed using progressive levels of exposure to living in a cold house with transition probabilities and sojourn time calculated for each model. Hazard ratios will be used to quantify the degree to which risk of *living in a cold house,* impacts on ARI in children, with children at *least risk of living in a cold house* used as the reference population. Hazard ratios (Cox proportional hazard) and associated confidence intervals will be calculated using the survival package, coxph and mstate extensions – to handle transition between multiple states.

Outputs

Outputs will be obtained using HMM$out_stan() and inspected using as.matrix to inspect posterior distribution as well as using density plots rstan::stan_dens().

HMM$viterbi() and HMM$plot_ts() will be used to visualise the distribution of the states over. HMM$plot() will be used to investigate transition probabilities between states. Variance will be reported in standard deviations using confidence intervals for plots.

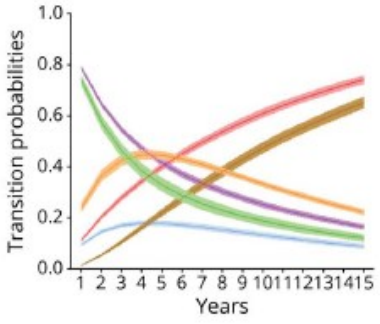
Hazard ratio's will be outputted as a table.

Visualisation

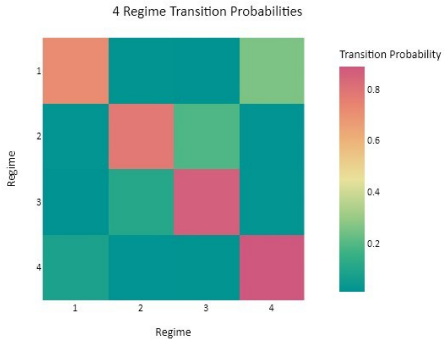
Visualisation will be conducted using bayesplot and ggplot2.

Splines: the transition probabilities listed will be mapped against the time variable (age) as B-Splines using splines and ggplot2:

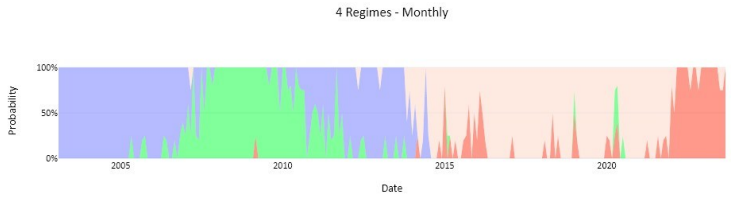
- Healthy -> Sick
- Healthy -> Very Sick
- Sick -> Healthy
- Sick-> Very Sick
- Very Sick -> Sick
- Very Sick -> Healthy



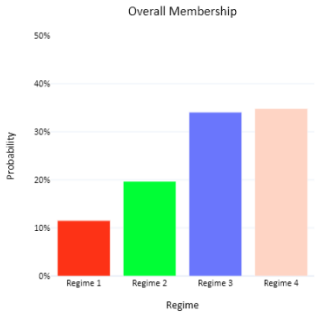
Heatmap: the transition probability for the whole follow-up period visualised as a heatmap using geom_tiles, with a colour gradient used to illustrate most likely transitions.

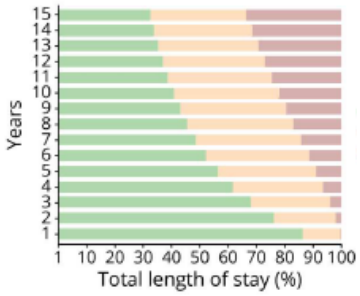


Proportional graph: proportion of children in each state at each timepoint will be categorized using the Viterbi output and plotted using geom_area.



Bar graphs: Mean sojourn time in each state across the whole follow-up period will be plotted using geom_histogram or geom_bar.





Appendix 1: Relative poverty line AHC Scotland.

To calculate exposure risk: *living in a cold house,* base year 2025, rate calculated from 2022-2025 IFS statistic and adjusted for inflation.

Year	Adjustment	Annual adjustment	Value
2012	-30.6%	-2.69%	£14,263
2013	-28.8%	-2.70%	£14,629
2014	-27.8%	-2.81%	£14,843
2015	-27.8%	-3.97%	£14.849
2016	-27.3%	-3.33%	£14,947
2017	-25.3%	-3.41%	£15,348
2018	-23.5%	-3.54%	£15,728
2019	-22.1%	-3.82%	£16,010
2020	-21.4%	-4.36%	£16,146
2021	-19.4%	-4.77%	£16,564
2022	-12.1%	-3.70%	£18,071
2023	-4.5%	-1.89%	£19,630
2024	-3.3%	-2.34%	£19,876
2025	/	/	£20,555

Appendix 2: Definitions of Fuel Poverty used in the UK

2M: Isherwood and Hancock (1979), households that spend more than twice the median proportion of income (2M) on fuel, light and power. Generally measured after housing costs.

>10%: Broadbent (1991) calculated the 2M energy poverty threshold after housing costs to be equivalent to spending more than 10% of household income on fuel and power.

>10% (required): UK Fuel Poverty Strategy (2001) quantified this as being where a household would *need to spend* more than 10% of their income to achieve a standard temperature of at least 21°c in living rooms and 18° in bedrooms.

LIHC: Robinson, Bouzarovski, & Lindley, (2018) the household's required energy costs exceed the national median level for similar households, and if the household were to spend that required amount, its remaining residual income would fall below the official poverty line (typically defined as less than 60% of the median equivalised income).

LILEE: Department for Energy Security and Net Zero, (2024) the household is living in a property with a fuel poverty energy efficiency rating of band D or below and should they spend the required amount to heat the property, they are left with a residual income below the official poverty line.

Appendix 3: uZero Fuel Poverty measure

fuel_poverty_score_oa	A score from 0-100 predicting the relative potential benefit of home improvements in reducing fuel poverty in each census output area. The output areas likely to benefit the most (of all the areas where the score is calculated) will have a score of 100. The output areas with the least potential impact on fuel poverty will have a score of 0. The score is calculated as the min-max normalised mean value of the following features after they are standardised: percentage_uc_population_oa; energy_efficiency_c_gap_mean_oa; percentage_devices_fuel_poor_gas_smoothed_lsoa; percentage_devices_fuel_poor_electricity_smoothed_lsoa. (Standardisation subtracts the mean from all values of a feature and divides by the standard deviation.) **UPDATE: Slight change of inputs: percentage_uc_population_oa now uses the number of domestic UPRNs (num_uprns_oa) instead of working-age population estimates.** **Note that the header has changed from "combined_fuel_poverty_score_v3" to "fuel_poverty_score".**